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CO electrolysers with 51% energy efficiency towards C_{2+} using porous separators

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Supplementary Note 1 | One-dimensional COMSOL simulation

The simulation of cation distributions was performed in the COMSOL Multiphysics package based on the finite-element-method solver. Generalised modified Poisson-Nernst-Plank (GMPNP) equations without the homogeneous reaction term were applied to calculate the cations distribution, E-field, and pH at the electrode-electrolyte interface. The GMPNP equations used to model mass transport of electrolyte species are given by[1,2]:

$$\frac{\partial C_i}{\partial t} = -\nabla \cdot \boldsymbol{J_i} \qquad \textbf{(Equation S1)}$$

$$\boldsymbol{J_i} = D_i \nabla C_i - \frac{D_i C_i z_i F}{RT} \nabla \phi - \frac{D_i C_i N_A a_i^3 \sum_i \nabla C_i}{1 - \sum_i a_i^3 C_i} \qquad \textbf{(Equation S2)}$$

$$\nabla \cdot (\varepsilon_0 \varepsilon_r \nabla \phi) = -\rho_f = -F \sum_i z_i C_i \qquad \textbf{(Equation S3)}$$

where C_i is the concentration of the ith species, $\boldsymbol{J_i}$ is the flux for the ith species, D_i is the diffusion constant of the ith species, a_i is the solvation size of the ith species, z_i is the valence charge of the ith species, F is the Faradaic constant, R is the ideal gas constant, T is the temperature, N_A is the Avogadro constant, ε_0 is the vacuum permittivity, ε_r is the relative permittivity of the solution. As defined in Gouy-Chapman-Stern (GCS) model, the OHP is the closet proximity of the ions to the electrode surface. Therefore, the Stern layer contains no charged species but water. Thus, the charge density at OHP, σ_{OHP}, is given by:

$$\sigma_{OHP} = \varepsilon_0 \varepsilon_r \left.\frac{d\phi}{dx}\right|_{x_{OHP}} = \frac{\varepsilon_0 \varepsilon_{Stern} (\phi_M - \phi_{OHP})}{d_{Stern}} \qquad \textbf{(Equation S4)}$$

where ε_{Stern} and d_{Stern} denote the relative permittivity and thickness of the Stern layer, respectively, ϕ_M and ϕ_{OHP} are the potentials at the electrode surface and OHP, respectively. Equation S1 specified the relationship between the potential (ϕ) and the potential gradient ($\frac{d\phi}{dx}$) at the OHP.

To account for the E-field dependence of the dielectric permittivity in the diffuse layer, the Booth model, equation (S4), is employed[3–5].

$$\varepsilon_r(E;\beta) = n^2 + (\varepsilon_r(0) - n^2)\frac{3}{\beta E}\left[\coth(\beta E) - \frac{1}{\beta E}\right] \qquad \textbf{(Equation S5)}$$

In equation S4, $E = |-\nabla\phi(r)|$ is the local E-field strength, n is the electrolyte refractive index, and β is a parameter given by:[6]

$$\beta = \frac{5\mu}{2kT}(n^2 + 2) \qquad \textbf{(Equation S6)}$$

where μ is the dipole moment of the water molecule.

Furthermore, to account for the effect of the E-field on dielectric permittivity in the stern layer, we extracted the potential-dependent value from equation (S7)-(S8) as suggested by Johnson et al.[1]:

$$C_{Stern} = \frac{\varepsilon_0 \varepsilon_{Stern}}{x_{Stern}} \qquad \textbf{(Equation S7)}$$

$$C_{Stern} = -1.8923\, V^3 - 4.4698\, V^2 + 2.0741 V + 26.323 \qquad \textbf{(Equation S8)}$$

Even though Equation (S8) being derived from experiments utilizing Na electrolyte, it provides a more plausible estimation of ε_{Stern} when compared with the scenario of presuming ε_{Stern} equals ε_e.

To simulate the mass transport in the porous medium (AEM), the effective diffusivity was considered in these simulation domains with the correction of porosity and tortuosity using the Bruggeman relationship[7,8]:

$$\mathrm{D}_{e,i} = \frac{\varepsilon}{\tau}\mathrm{D}_i = \varepsilon^{3/2}\mathrm{D}_i \qquad \textbf{(Equation S9)}$$

Moreover, positive charge with constant concentration was added to AEM domain to simply mimic the fixed charge in AEM and the corresponding ion selective transportation effect.
The heterogeneous reactions (S10) and (S11) are considered to be occurring at the planar electrode surface in the COR process:

$$2CO\,(aq) + 6H_2O + 8e^- \rightleftarrows C_2H_4(\mathrm{g}) + 8OH^- \qquad \textbf{(Equation S10)}$$

$$2H_2O + 2e^- \rightleftarrows H_2(\mathrm{g}) + 2OH^- \qquad \textbf{(Equation S11)}$$

We note that at Angstrom-level-scales relevant to the electrical double layer, continuum models such as General modified Poisson–Nernst–Planck (GMPNP) model may not accurately reflect discrete solvent effects and specific ion-solvent interactions. Thus, the actual cation distribution may exhibit greater complexity than predicted here. Nevertheless, our continuum simulations remain useful in identifying general trends and comparing ion accumulation behaviors across different configurations.

Supplementary Table 1 | Diffusion-coefficients in m^2s^{-1}

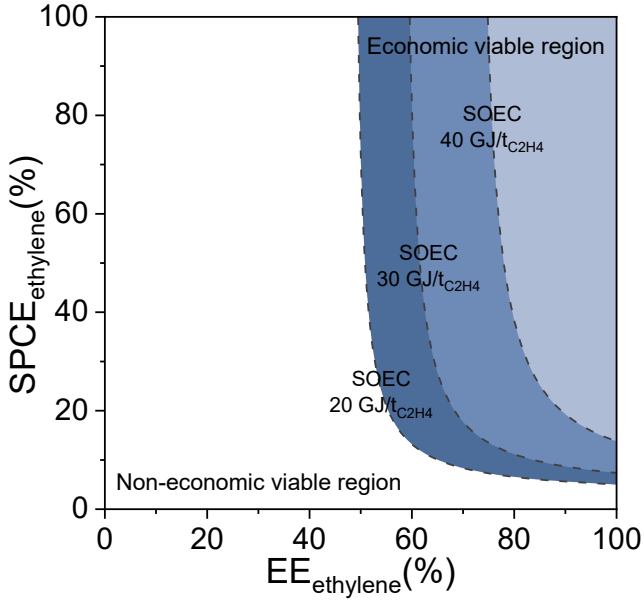
Constant	Value	Reference
D_{H^+}	9.311×10^{-9}	[2]
D_{Na^+}	1.334×10^{-9}	[2]
D_{OH^-}	5.273×10^{-9}	[2]
D_{CO} *	1.91×10^{-9}	[2]

*The diffusion coefficient of CO is assumed to be similar to CO_2.

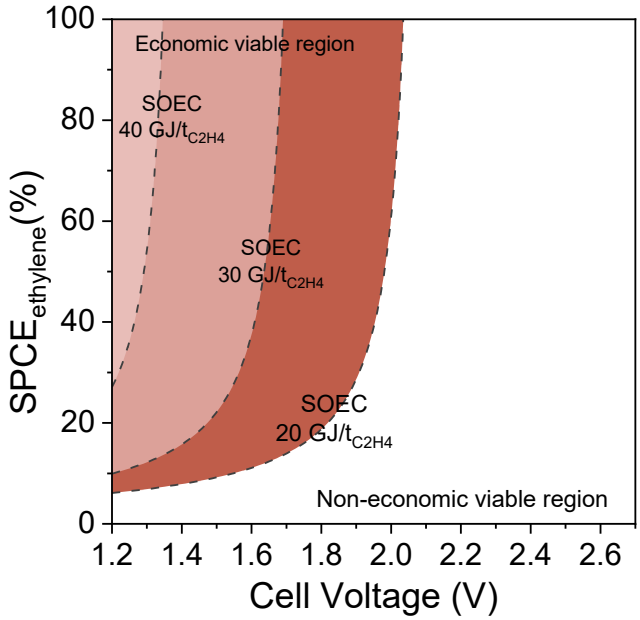
Supplementary Table 2 | Solvation size in *m*:

Constant	Value	Reference
a_{H^+}	0.56×10^{-9}	2
a_{Na^+}	0.716×10^{-9}	2
a_{OH^-}	0.6×10^{-9}	2
a_{CO} *	0.115×10^{-9}	

*The solvated size of CO is assumed to be the C−O bond distance.



Supplementary Fig. 1 | Techno-economic analysis (TEA) of energy efficiency requirement for economic viable COR. C_2H_4 was chosen as the model molecule with an idealized FE of 95%. The economic viable region corresponds to energy intensity less than 80 GJ ton_{C2H4}^{-1} which includes costs for both the CO_2-to-CO SOEC and the CO-to-C_2H_4 COR steps. The 80 GJ ton_{C2H4}^{-1} target corresponds to a total process efficiency of ~55% given the lower heating value of C_2H_4 to be 45 GJ ton^{-1}. The three dash lines correspond to the EE_{C2H4} and $SPCE_{C2H4}$ combination required to achieve the 80 GJ ton_{C2H4}^{-1} target if the SOEC step is 20 GJ ton_{C2H4}^{-1} (or 10 GJ ton_{CO}^{-1}), 30 GJ ton_{C2H4}^{-1} (or 15 GJ ton_{CO}^{-1}), and 40 GJ ton_{C2H4}^{-1} (or 20 GJ ton_{CO}^{-1}), respectively.



Supplementary Fig. 2 | TEA of cell voltage requirement for economic viable COR. C_2H_4 was chosen as the model molecule with an idealized FE of 95%. The economic viable region corresponds to energy intensity less than 80 GJ ton_{C2H4}^{-1} which includes costs for both the CO_2-to-CO SOEC and the CO-to-C_2H_4 COR steps. The 80 GJ ton_{C2H4}^{-1} target corresponds to a total process efficiency of ~55% given the lower heating value of C_2H_4 to be 45 GJ ton^{-1}. The three dash lines correspond to the cell voltage and $SPCE_{C2H4}$ combination required to achieve the 80 GJ ton_{C2H4}^{-1} target if the SOEC step is 20 GJ ton_{C2H4}^{-1} (or 10 GJ ton_{CO}^{-1}), 30 GJ ton_{C2H4}^{-1} (or 15 GJ ton_{CO}^{-1}), and 40 GJ ton_{C2H4}^{-1} (or 20 GJ ton_{CO}^{-1}), respectively.

Supplementary Note 2 | TEA of tandem CO_2-CO and CO-to-C_{2+} system.

Here we provide a brief description of TEA model. This TEA model follows method as described previously[9]. As each C_{2+} product has different thermodynamic properties, we consider C_2H_4 as a model molecule for the analysis.

We consider economic viable C_2H_4 electrosynthesis based on a total process efficiency of ~55% (from CO_2, C_2H_4 lower heating value = 45 GJ ton^{-1}) which corresponds to a final energy intensity target of 80 GJ ton_{C2H4}^{-1} [10]. Meeting this target requires low energy intensities for both steps (CO_2-to-CO and CO_2-to-C_2H_4).

For the first SOEC CO_2-to-CO step, we consider three scenarios: 1) 20 GJ ton_{C2H4}^{-1} (or 10 GJ ton_{CO}^{-1}); 2) 30 GJ ton_{C2H4}^{-1} (or 15 GJ ton_{CO}^{-1}); and 40 GJ ton_{C2H4}^{-1} (or 20 GJ ton_{CO}^{-1}). The SOEC energy costs include electricity input, heat input and product (CO_2/CO) separation. Unreacted CO, upon separation with CO_2, is recycled to the inlet of the SOEC. For the COR step, we consider an idealized C_2H_4 FE of 95% (corresponds to >95% FE_{C2+} in this work) and H_2 as the only by-product (5% FE_{H2}). A pressure swing adsorption (PSA) module is considered at the downstream of the cathode outlet to separate C_2H_4 from unreacted CO and H_2 by-product. The unreacted CO is returned to the cathode inlet for utilization. Calculation details are included in Supplementary Note 2.

Cases with higher SOEC input energy demand greater performance of the COR step. All cases demand a combination of high SPCE and EE that is beyond the state of the art. The results show the EE_{C2H4} and $SPCE_{C2H4}$ combination required to achieve the 80 GJ ton_{C2H4}^{-1} target. The results show that achieving EE_{C2H4} > ~50% (cell voltage < 2 V) and $SPCE_{C2H4}$ > ~60% enables the 80 GJ ton_{C2H4}^{-1} target if the SOEC energy input is kept low at 20 GJ ton_{C2H4}^{-1}. This TEA model provides a guideline for EE and voltage target for C_{2+} synthesis via COR.

Supplementary Note 3 | Detail calculation of the TEA used for the tandem CO_2-to-CO and CO-to-C_{2+} system.

COR energy cost. We calculate the production rate (mol s^{-1}) of C_2H_4 based on a production capacity of 100 tons per day.

$$Production\ rate\ \left[\frac{mol}{s}\right] = \frac{Production\left[\frac{g}{day}\right]}{molecular\ weight_{C_2H_4}\left[\frac{g}{mol}\right] \times 86400\left[\frac{s}{day}\right]}$$

$$Production\ rate\ \left[\frac{mol}{s}\right] = \frac{100 \times \frac{10^6 g}{day}}{\frac{28\ g}{mol} \times \frac{86400 s}{day}} = 41.34\ \frac{mol}{s}$$

The total current required to reach this production rate is:

$$Total\ current\ required\ [A] = \frac{production\ rate\left[\frac{mol}{s}\right] \times No.e^{-}\ transferred \times Faraday's\ constant}{FE_{C_2H_4}}$$

$$Total\ current\ required\ [A] = \frac{41.34\frac{mol}{s} \times 8 \times 96485\frac{sA}{mol}}{95\%} = 33\ 588\ 968\ \ A$$

The power consumption is calculated by multiplying the current with the full-cell voltage (taking 2 V as an example).

$$Power\ Consumed\ [W] = Total\ current\ needed\ [A] \times Cell\ voltage\ [V] = 33\ 588\ 968\ \ A\ \times 2\ V = 67\ 178\ kW$$

We find the total energy input to produce 100 tons of C_2H_4 per day:

$$Electricity\ Energy\ Requirement\ \left[\frac{GJ}{ton_{C_2H_4}}\right] = \frac{Power\ Consumed\ [W] \times 24[h]}{daily\ production\ [ton]} = \frac{67\ 178\ kW \times 24\ h \times 0.0036\frac{GJ}{kWh}}{100\ ton} = \mathbf{58.0\ \frac{GJ}{ton_{C_2H_4}}}$$

Downstream gas separation energy cost. A pressure swing adsorption (PSA) separation module is used to calculate the energy cost associated with the separation of C_2H_4, H_2, CO from the cathodic downstream.

$$PSA\ operating\ energy\ \left[\frac{kWh}{ton_{C_2H_4}}\right] = 0.25\ \frac{kWh}{m^3} \times flow\ rate\left[\frac{m^3}{h}\right] \times 24\frac{h}{day}$$

The flow rate of each product at the cathodic stream is estimated, assuming operation under the ideal conditions.

$$Output\ C_2H_4\ flow\ rate\ \left[\frac{m^3}{h}\right] = \frac{100 \times 10^6\ \frac{g}{day} \times 8.314\ Jmol^{-1}K^{-1} \times 298K}{28\frac{g}{mol} \times 101\ 300\ Pa \times 24\frac{h}{day}} = 3\ 640\ \frac{m^3}{h}$$

We then estimate the flow rates of unreacted CO at the cathodic stream. The flow rate of CO at the cathodic stream is calculated by using the SPCE. We then calculate the output unreacted CO flow rate using a SPCE of 50%:

$$Output\ CO\ flow\ rate\ \left[\frac{m^3}{h}\right] = CO\ requirement\ \left[\frac{mol}{s}\right] \times 3600\left[\frac{s}{h}\right] \times \frac{8.314\frac{J}{mol \times K} \times 298K}{101.3 \times 10^3 Pa} \times \left(\frac{1 - SPCE}{SPCE}\right)$$

$$= 82.68\frac{mol}{s} \times 3600\frac{s}{h} \times \frac{8.314\frac{J}{mol \times K} \times 298K}{101.3 \times 10^3 Pa} \times \frac{1 - 0.6}{0.6} = 4\ 853\ \frac{m^3}{h}$$

We then calculate flow rate for H_2:

$$Current\ toward\ H_2[A] = Total\ current\ needed[A] \times FE_{H_2}$$

$$Current\ toward\ H_2[A] = 33\ 588\ 968\ \ \times 5\% = 1\ 679\ 448\ A$$

$$H_2\ production\ \left[\frac{mol}{h}\right] = \frac{Current\ toward\ H_2[A] \times 3600\frac{s}{h}}{2\frac{electrons}{H_2\ product} \times Faraday's\ constant}$$

$$H_2\ production\ \left[\frac{mol}{h}\right] = \frac{1\ 679\ 448\ \ A \times 3600\frac{s}{hour}}{2\frac{electrons}{H_2\ product} \times 96485\frac{sA}{mol}} = 31\ 331\frac{mol}{h}$$

Assuming the produced H_2 as an ideal gas under standard conditions, the flow rate of H_2 can be calculated:

$$flow\ rate_{H_2}\left[\frac{m^3}{h}\right] = \frac{H_2\ production\left[\frac{mol}{h}\right] \times 8.314\frac{J}{mol \times K} \times 298K}{101.3 \times 10^3 Pa}$$

$$flow\ rate_{H_2}\left[\frac{m^3}{h}\right] = \frac{31\ 311\frac{mol}{h} \times 8.314\frac{J}{mol \times K} \times 298K}{101.3 \times 10^3 Pa} = 7\ 66\ \frac{m^3}{h}$$

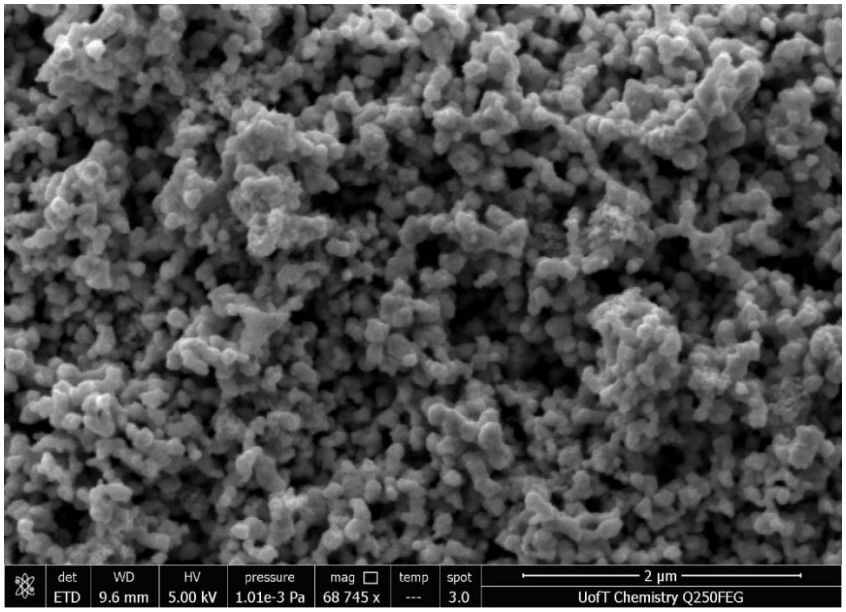
The total flow rate of the cathodic stream can then be calculated by adding the flow rates of C_2H_4, unreacted CO, and H_2:

$$flow\ rate\ \left[\frac{m^3}{h}\right] = (3\ 640 + 4\ 853 + 7\ 66)\frac{m^3}{h} = 9\ 259\frac{m^3}{h}$$

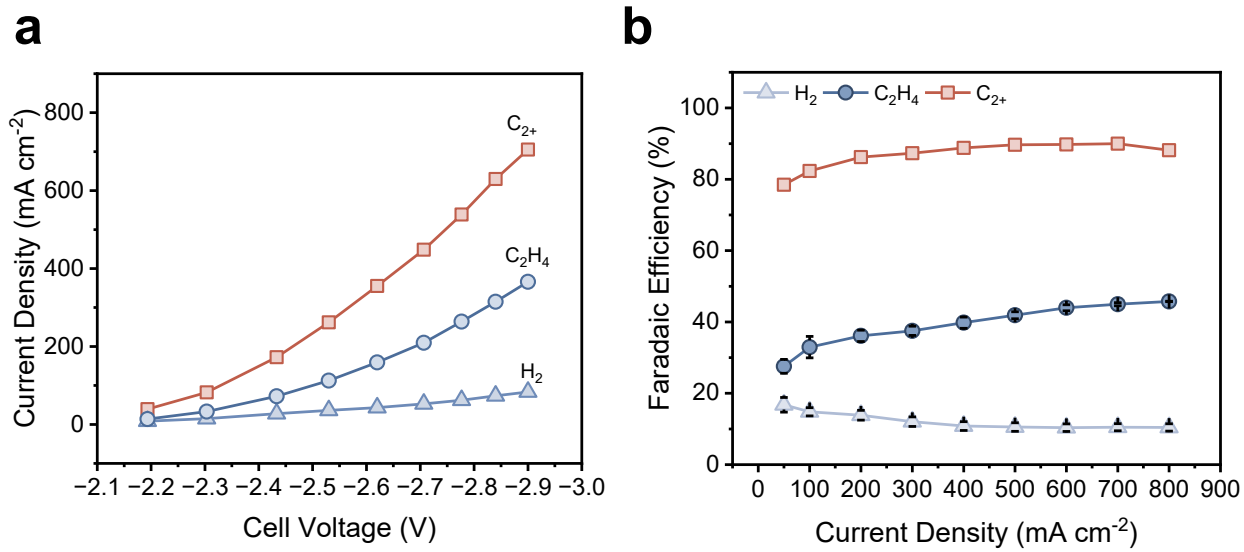
We can then calculate the total energy required to operate the PSA separation module:

$$PSA\ energy\ \left[\frac{GJ}{ton_{C_2H_4}}\right] = 0.25\ \frac{kWh}{m^3} \times 9\ 259\frac{m^3}{h} \times 24\frac{h}{day} \times \frac{0.0036\ GJ\ kWh^{-1}}{100\ ton_{C_2H_4}}$$

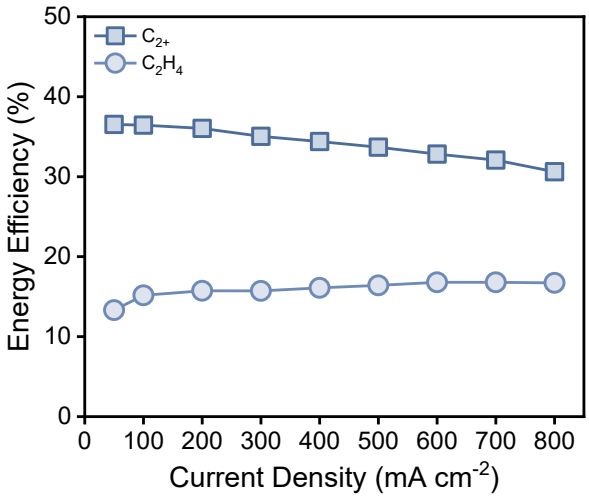
$$= 2.0\ \frac{GJ}{ton_{C_2H_4}}$$



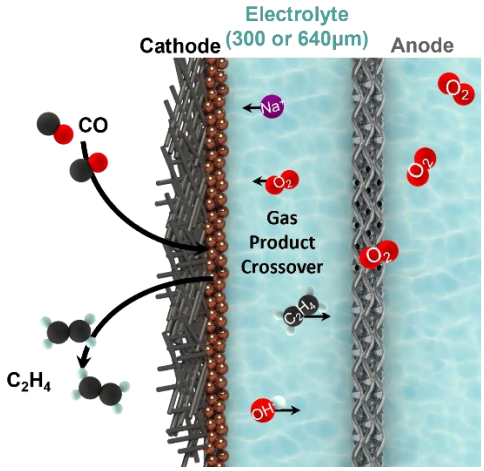
Supplementary Fig. 3 | Scanning electron microscopy (SEM) image of the Cu electrode.



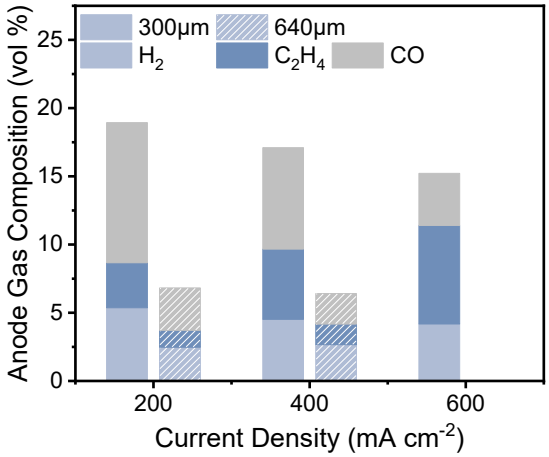
Supplementary Fig. 4 | COR performance of the MEA with an AEM. (a) H_2, C_2H_4 and C_{2+} partial current density. (b) H_2, C_2H_4 and C_{2+} FE. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).



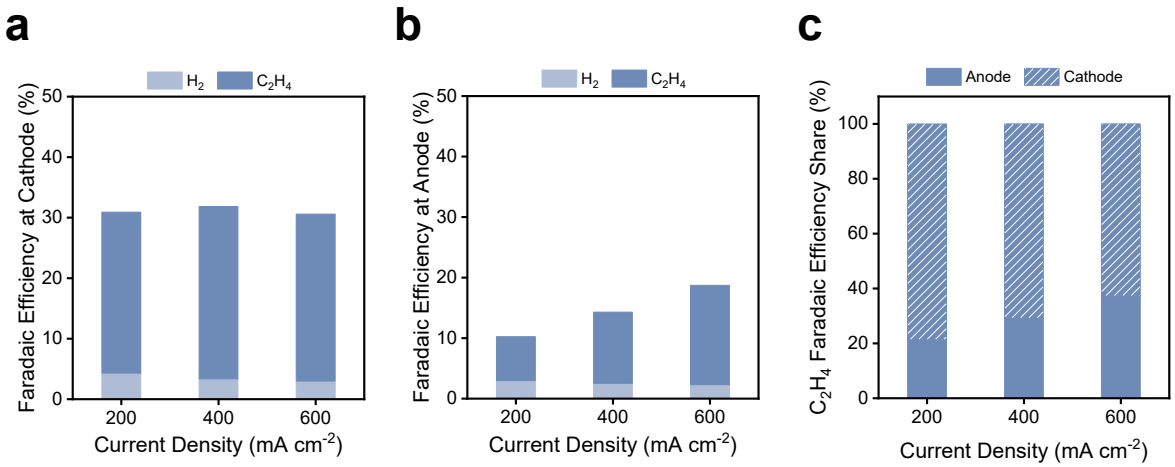
Supplementary Fig. 5 | Energy efficiency of C_2H_4 and C_{2+} products from CO of the MEA with an AEM.



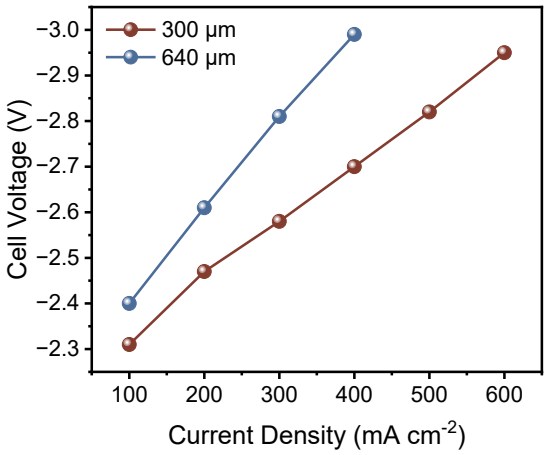
Supplementary Fig. 6 | Schematic the membrane-free cell.



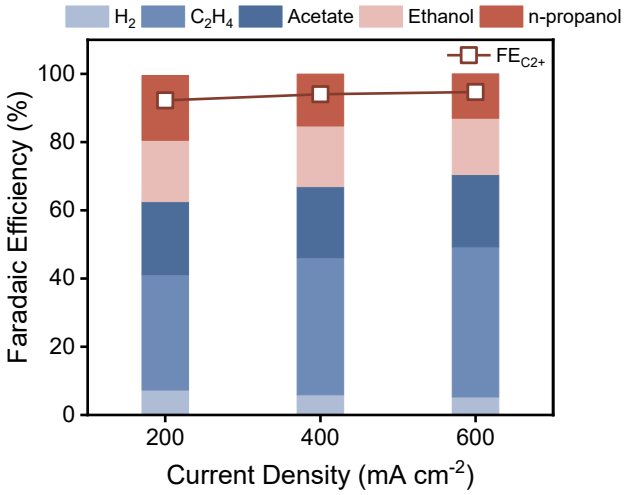
Supplementary Fig. 7 | Anode gas compositions at varying inter-electrode separation thicknesses in the membrane-free cell. The remaining anode gas consists of O_2 from the oxygen evolution reaction (OER). Separation thicknesses were defined under compression within the cell.



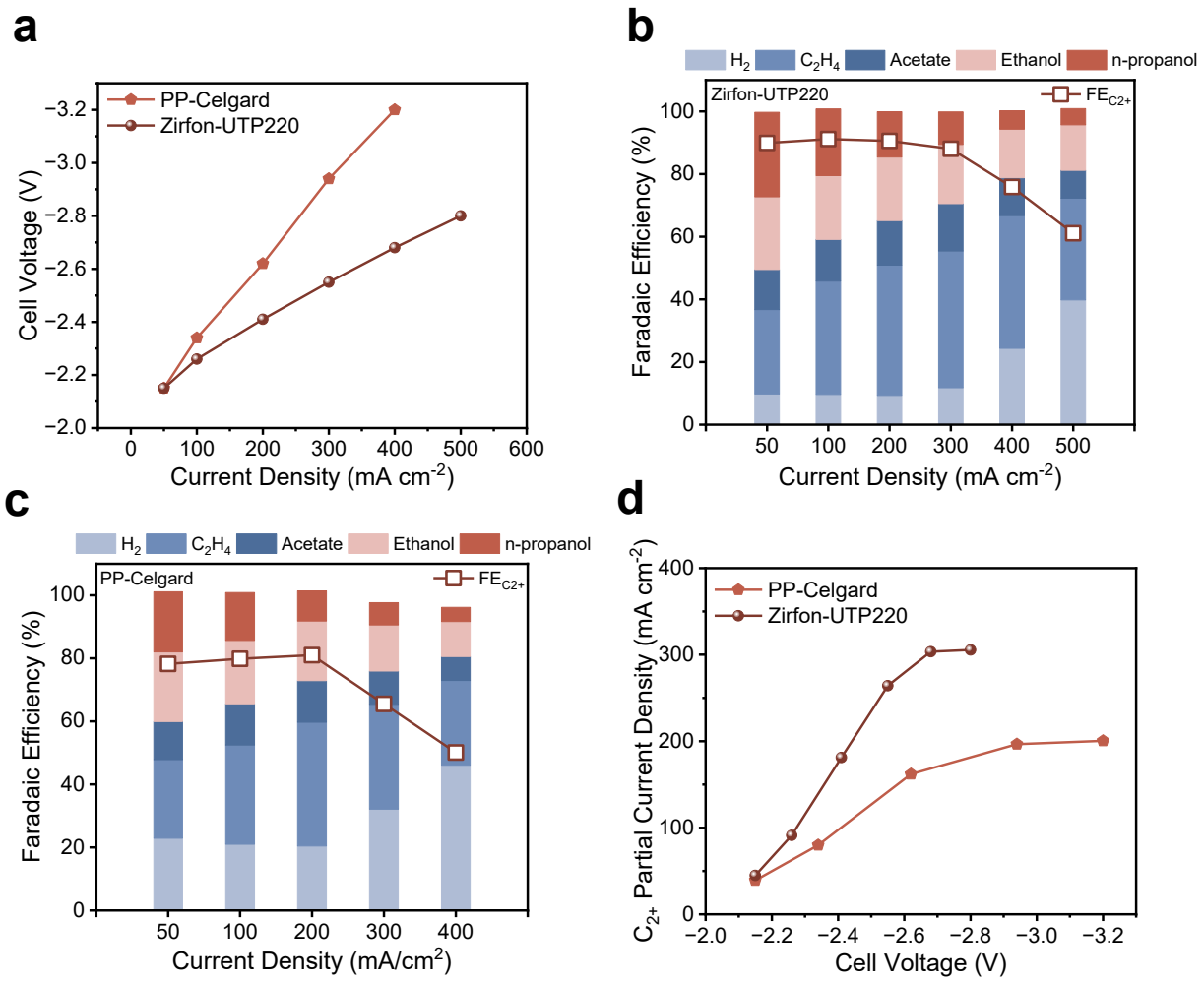
Supplementary Fig. 8 | Cathode gas product crossover at different current densities of the membrane-free cell. (a) COR gas product FE found at the cathode gas outlet. (b) COR gas product FE found at the anode gas outlet. (c) C_2H_4 FE shares at the cathode and anode.



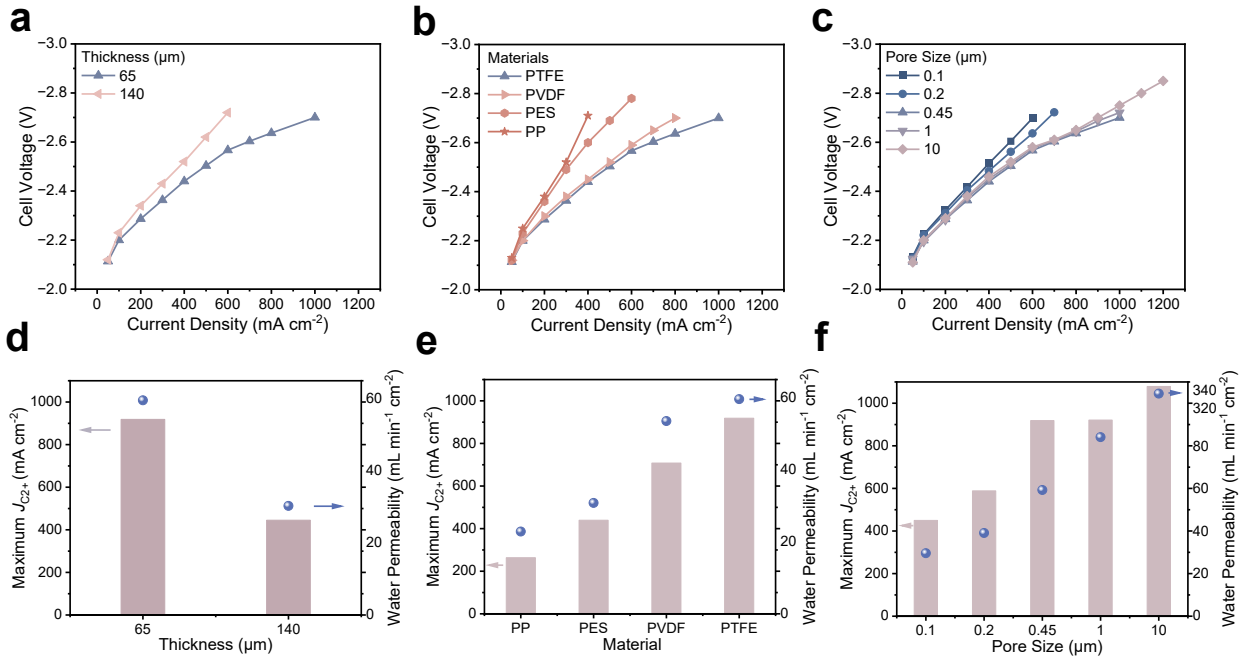
Supplementary Fig. 9 | Relationship between cell voltage and current density at varying inter-electrode separation thicknesses in the membrane-free cell.



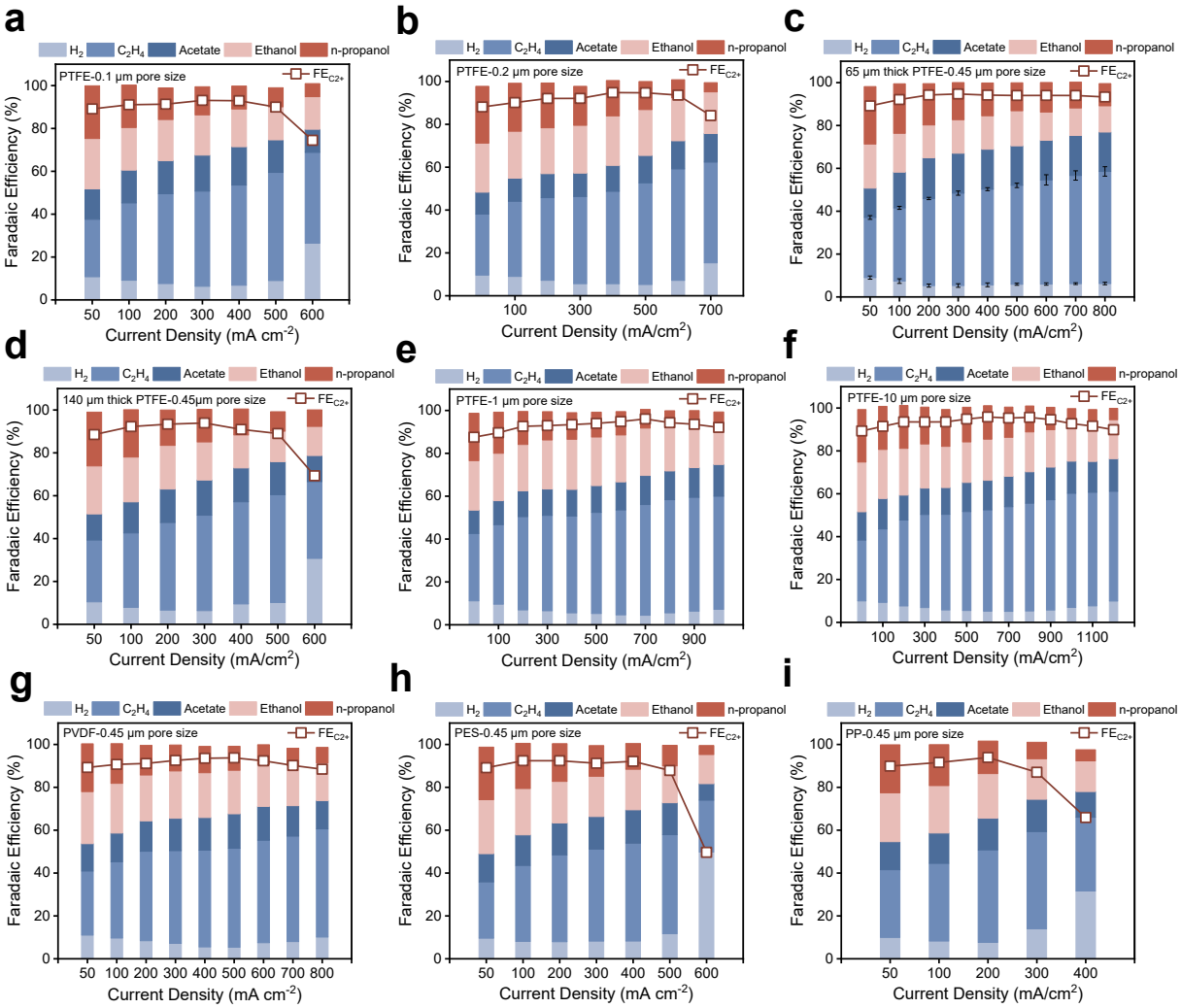
Supplementary Fig. 10 | Product distributions of the membrane-free cell at different current densities.



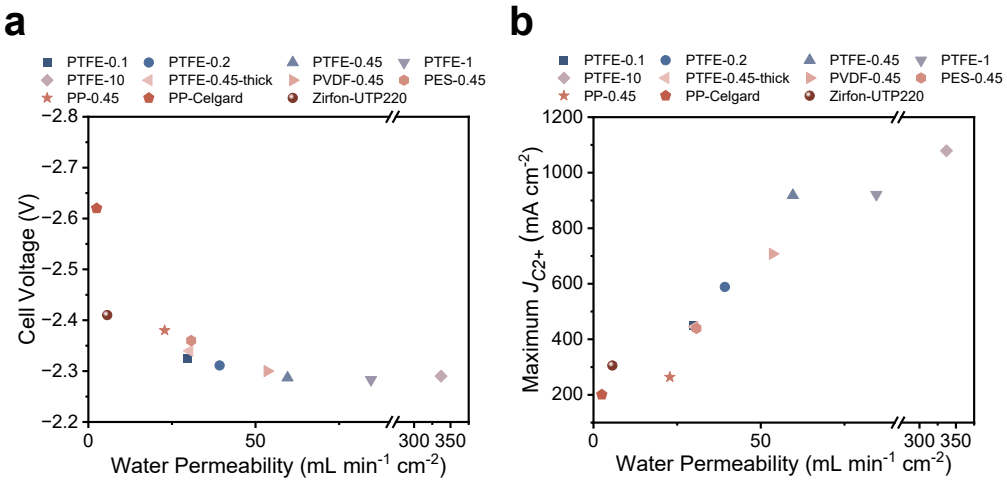
Supplementary Fig. 11 | COR performance with standard alkaline water electrolyzer separators – Zirfon and Celgard. (a) Relationship between cell voltage and current density. COR product distribution of (b) Zirfon-UTP220 and (c) PP-Celgard. (d) Relationship between C_{2+} partial current density and cell voltage. Detail properties of these separator materials are listed in Supplementary Table 3.



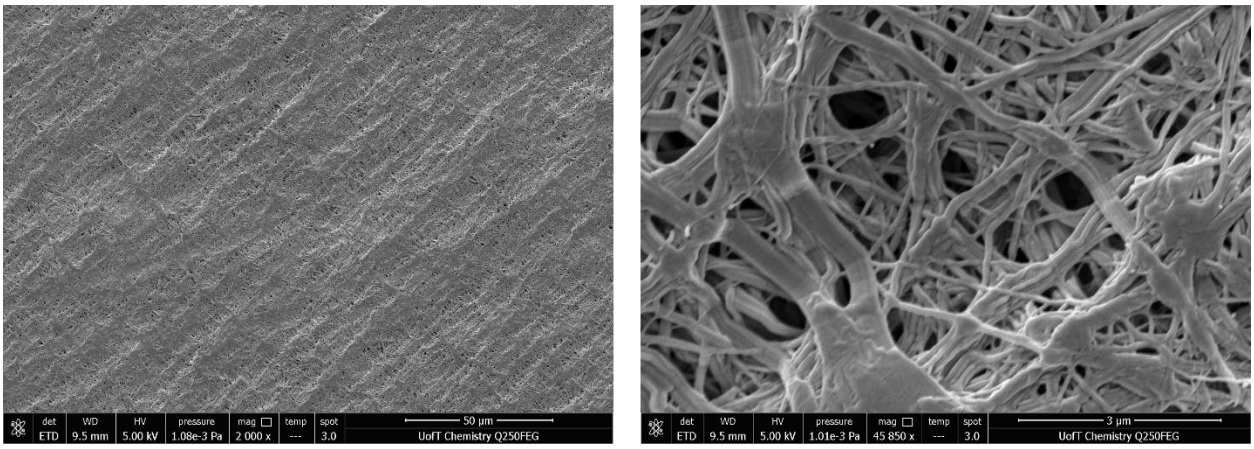
Supplementary Fig. 12 | COR performance with different separator materials. Relationship between cell voltage and current density comparing (a) different separator thicknesses; (b) different separator materials; (c) different separator pore sizes. Maximized C_{2+} partial current density achieved comparing (d) different separator thicknesses; (e) different separator materials; (f) different separator pore sizes.



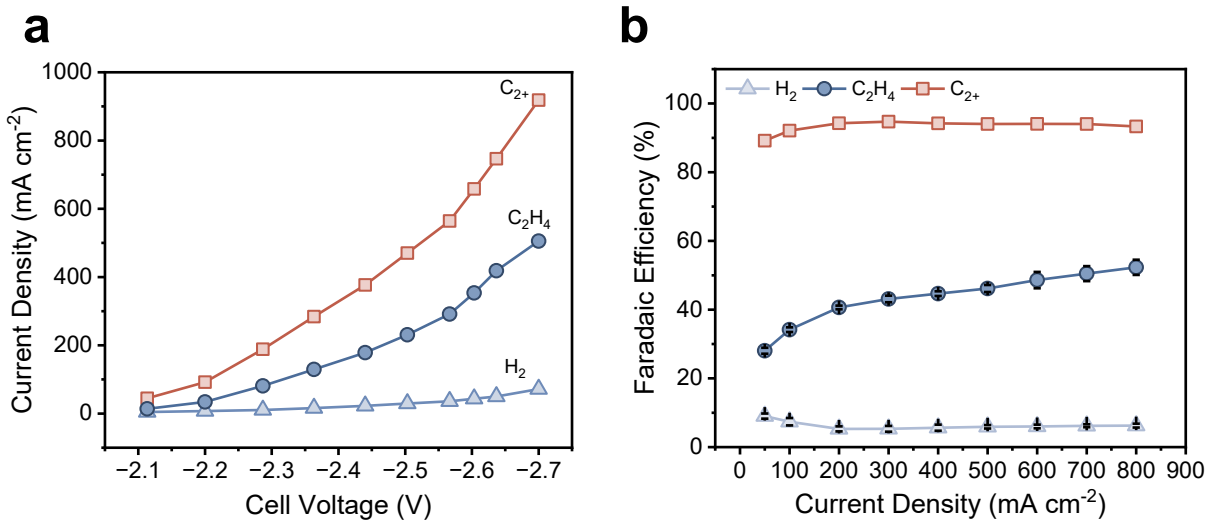
Supplementary Fig. 13 | COR product distribution with different separator materials. (a) PTFE-0.1, (b) PTFE-0.2, (c) PTFE-0.45-thin, (d) PTFE-0.45-thick, (e) PTFE-1, (f) PTFE-10, (g) PVDF-0.45, (h) PES-0.45, (i) PP-0.45. Detail properties of these separator materials are listed in Supplementary Table 3. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).



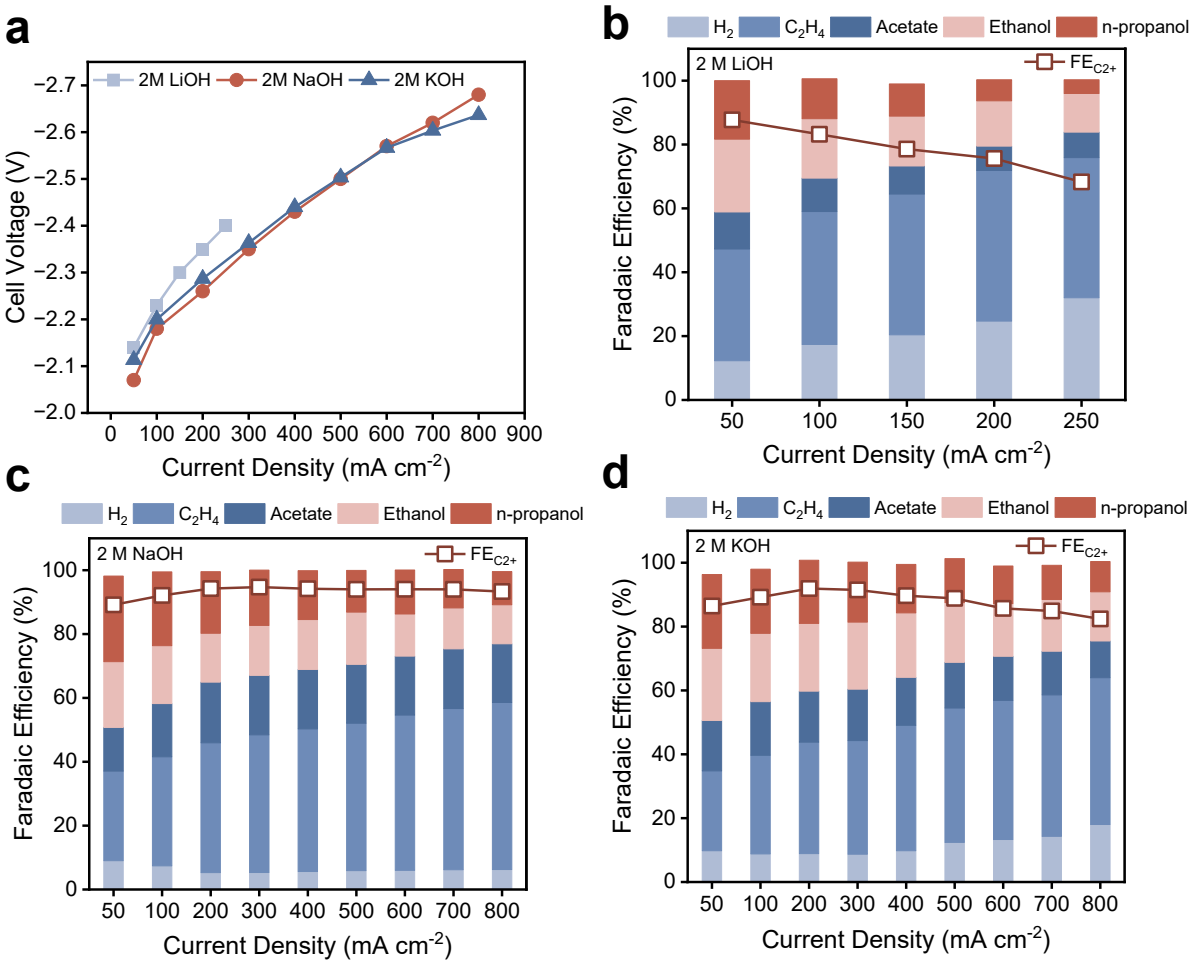
Supplementary Fig. 14 | Effect of separator water permeability on electrolyzer performance. (a) Relationship between cell voltage (at 200 mA cm^{-2}) and water permeability through the separator. (b) Relationship between maximum achieved C_{2+} partial current density with water permeability through the separator. Water permeabilities (mL min^{-1} cm^{-2}) were recorded by measuring the water flow rates through the separators with other side of the separators held under vacuum at a pressure of 0.75 bar. Separator materials with higher water permeabilities achieved higher maximum C_{2+} partial current density and lower voltage.



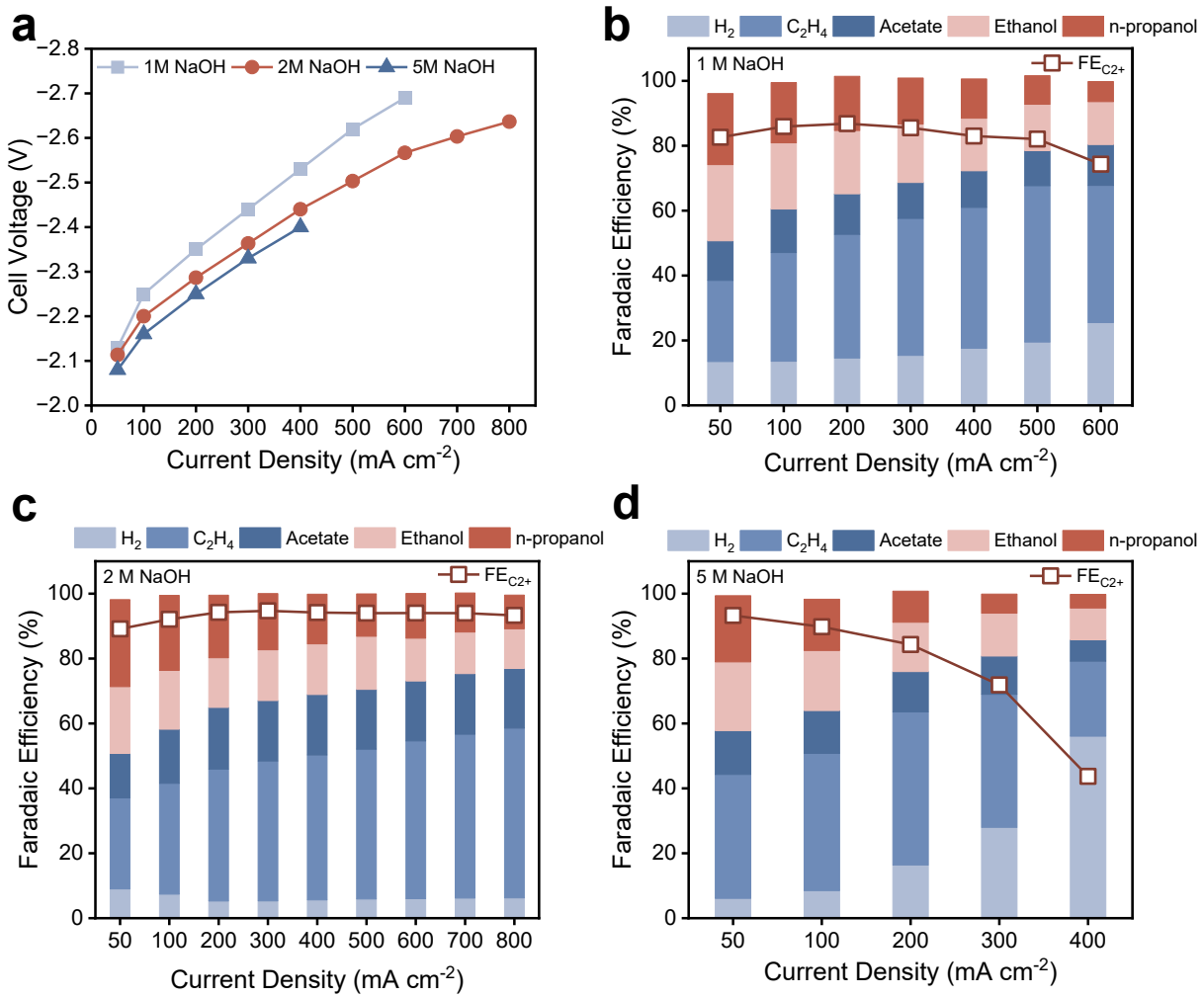
Supplementary Fig. 15 | SEM images of the porous hydrophilic polytetrafluoroethylene (PTFE) separator.



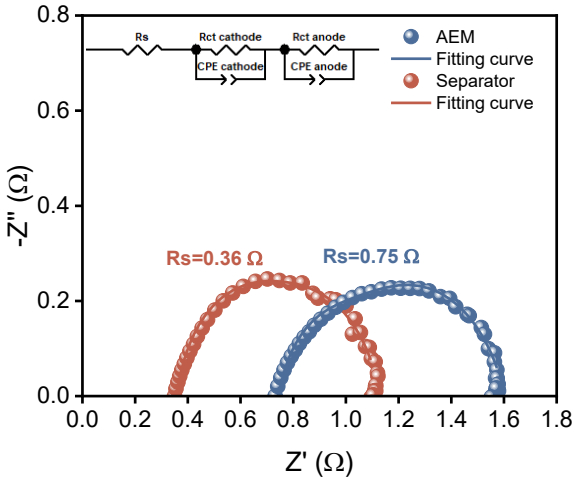
Supplementary Fig. 16 | COR performance of the MEA with a separator. (a) H_2, C_2H_4 and C_{2+} partial current density. (b) H_2, C_2H_4 and C_{2+} FE. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).



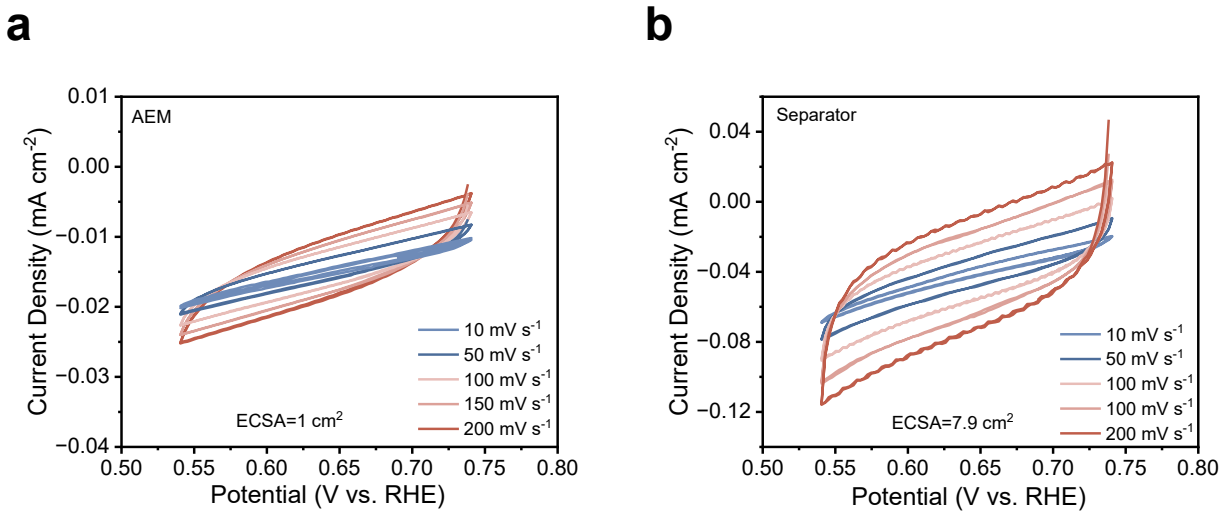
Supplementary Fig. 17 | COR performance of the MEA with a separator using different alkali hydroxides as the anolyte. Relationship between FE and current density for (a) 2 M LiOH, (b) 2 M NaOH, (c) 2 M KOH.



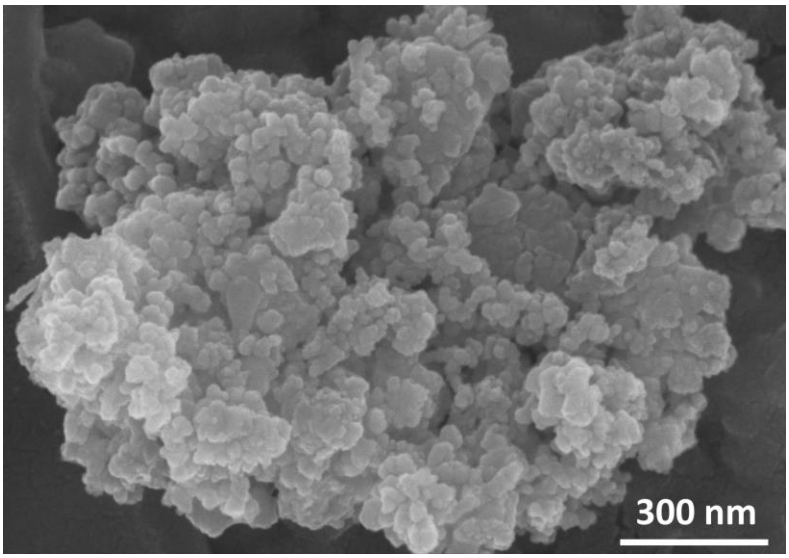
Supplementary Fig. 18 | COR performance the MEA with a separator in different anolyte concentrations. Relationship between FE and current density for (a) 1 M NaOH, (b) 2 M NaOH, (c) 5 M NaOH.



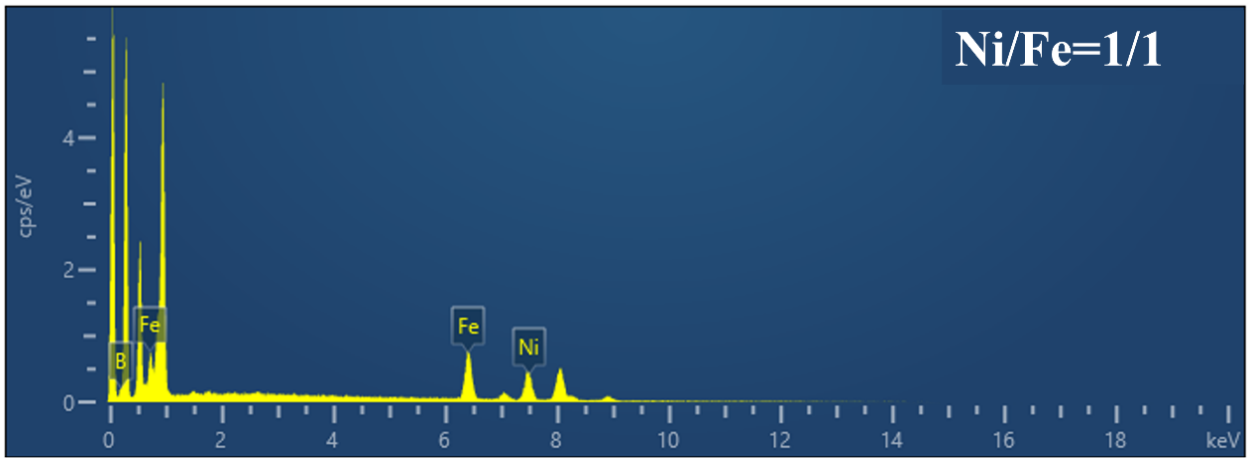
Supplementary Fig. 19 | Nyquist plot from full cell EIS of the MEA with an AEM and a separator, recorded at 200 mA cm^{-2}.



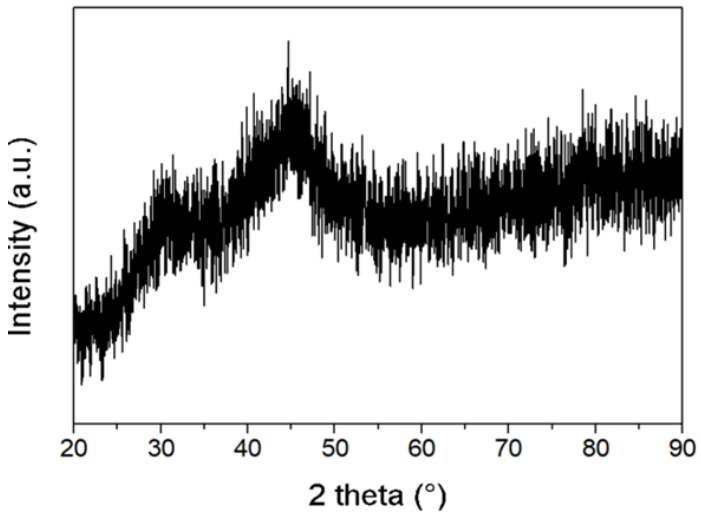
Supplementary Fig. 20 | Cyclic voltammograms for measuring the electrochemical surface area of the Cu catalyst in (a) the MEA with an AEM and (b) with a separator. A specific capacitance of 20 μF cm^{-2} is used for the Cu catalyst.[11]



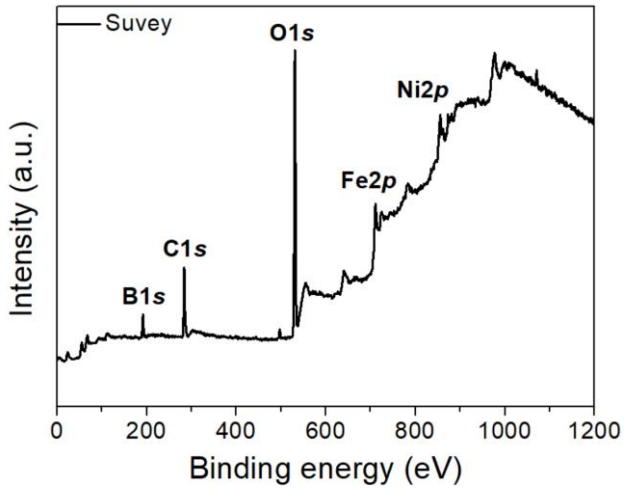
Supplementary Fig. 21 | SEM image of the NiFe-B anodic OER catalyst.



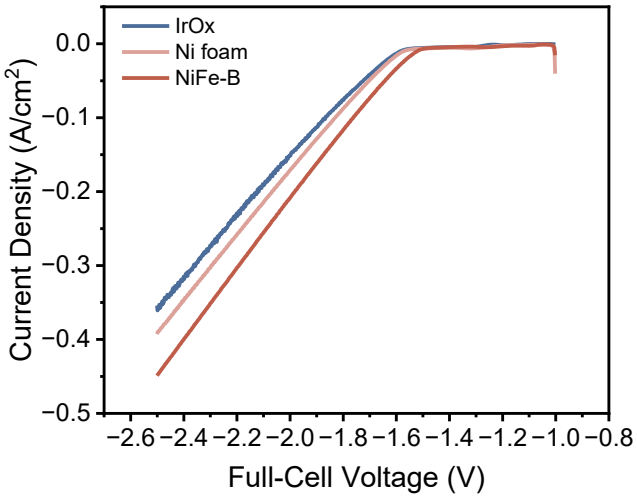
Supplementary Fig. 22 | Energy-dispersive X-ray spectroscopy (EDX) spectra of the NiFe-B catalyst.



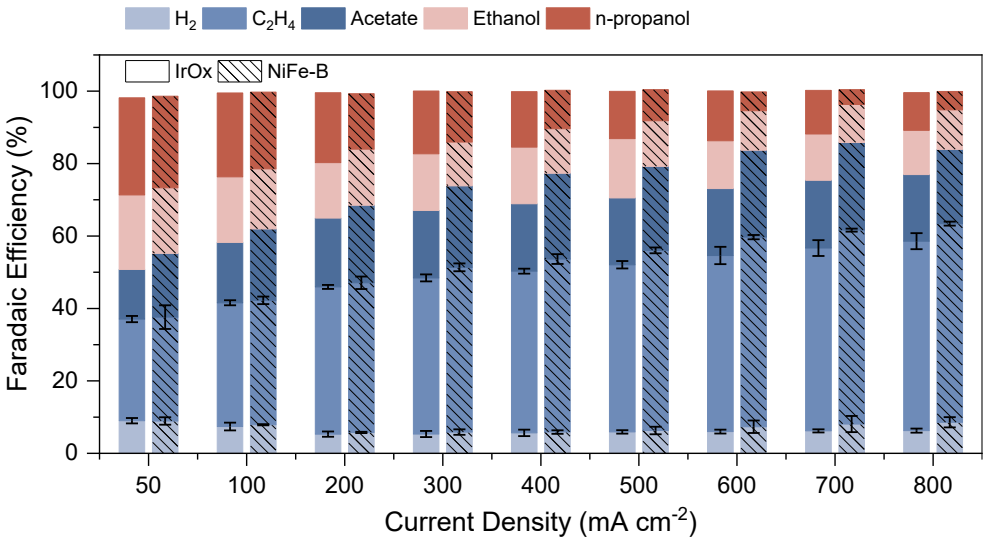
Supplementary Fig. 23 | X-ray diffraction (XRD) pattern of the NiFe-B catalyst. A broad XRD diffraction peak between 40º and 50º indicates low crystallinity in the catalyst.



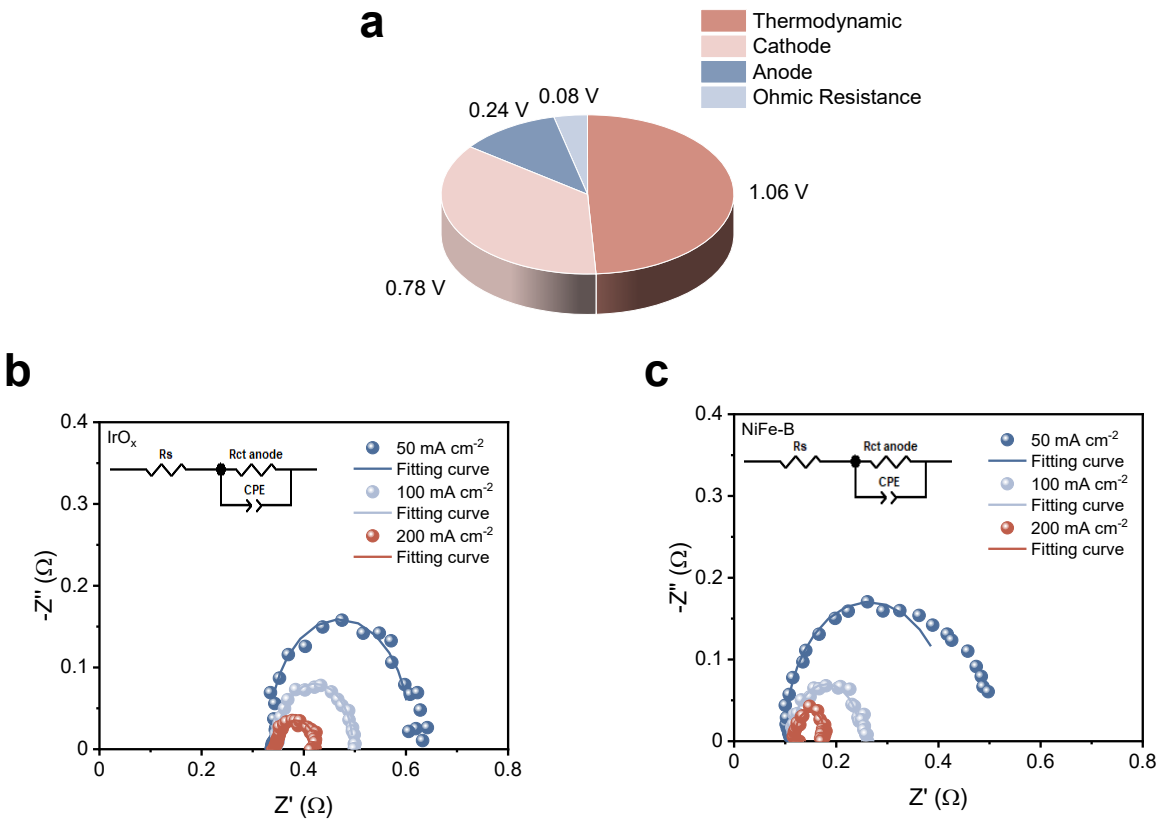
Supplementary Fig. 24 | The survey X-ray photoelectron spectroscopy (XPS) spectrum of the NiFe-B catalyst.



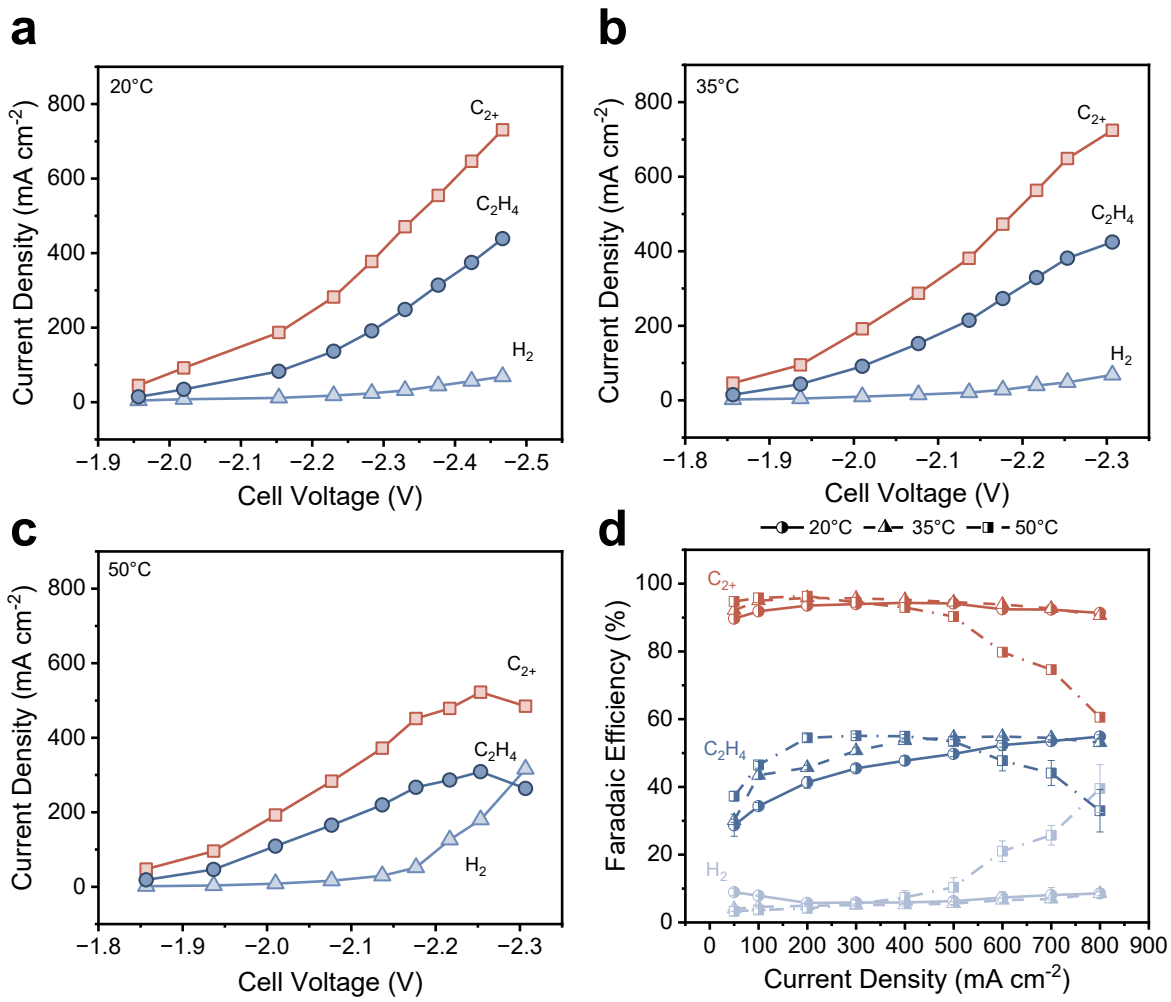
Supplementary Fig. 25 | Linear sweep voltammetry (LSV) of different OER catalysts in an MEA with a Cu cathode and an AEM. Scan rates were set at 10 mV s^{-1}. Cathode side was fed with Ar gas and 1 M NaOH was circulated at the anode.



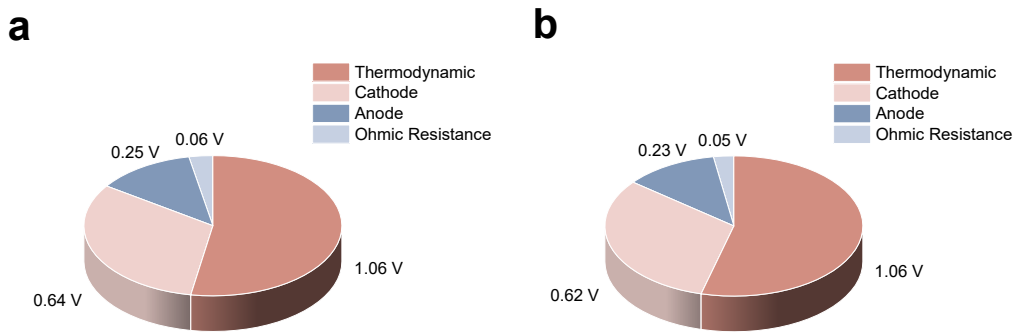
Supplementary Fig. 26 | COR product distribution with different OER catalysts in the MEA with a separator. The NiFe-B OER catalyst resulted in slightly higher acetate FE and lower ethanol and n-propanol FE which we attributed to the higher alcohol oxidation activity with the Ni-based catalysts. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).



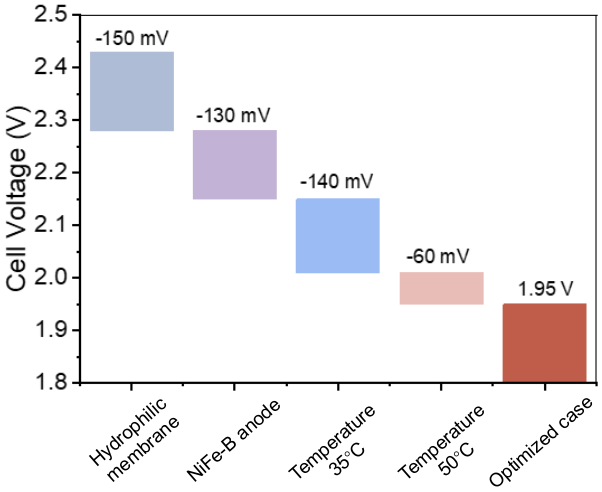
Supplementary Fig. 27 | Voltage breakdown of the system incorporating a separator and NiFe-B anode. (a) Resolved voltage profile of the MEA equipped with a separator and NiFe-B anode, recorded at 200 mA cm^{-2}. (b) Nyquist plot from half-cell EIS (anode vs. reference) of the in the MEA with a separator and IrOx anode. (c) Nyquist plot from half-cell EIS (anode vs. reference) of the in the MEA with a separator and NiFe-B anode.



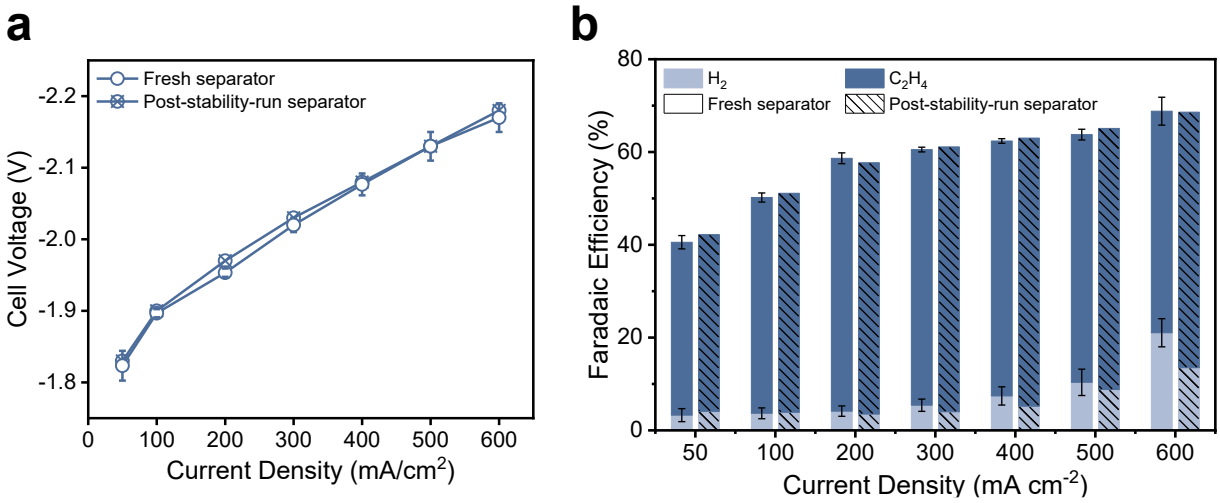
Supplementary Fig. 28 | COR performance of the MEA with a separator at different operating temperatures. H_2, C_2H_4 and C_{2+} partial current density at (a) 20°C, (b) 35°C, (c) 50°C. (d) H_2, C_2H_4 and C_{2+} FE at different operating temperatures. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).



Supplementary Fig. 29 | Voltage breakdown of the MEA with a separator at elevated operating temperatures (a) 35°C, (b) 50°C.



Supplementary Fig. 30 | Waterfall plot showing the voltages improvements observed at 200 mA cm^{-2}.



Supplementary Fig. 31 | COR performance of the post-stability-run separator. (a) Relationship between voltage and current density. (b) H_2 and C_2H_4 FE at different current density. Temperature was held at 50°C. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).

Supplementary Table 3 | Properties of different separator materials.

Material	Name	Pore size	Porosity	Thickness
Polytetrafluoroethylene (PTFE)	PTFE-0.1	0.1 μm	80%	30 μm
	PTFE-0.2	0.2 μm	80%	65 μm
	PTFE-0.45	0.45 μm	80%	65 μm
	PTFE-1	1 μm	80%	85 μm
	PTFE-10	10 μm	80%	85 μm
	PTFE-0.45-thick	0.45 μm	80%	140 μm
Polyvinylidene fluoride (PVDF)	PVDF-0.45	0.45 μm	70%	125 μm
Polyethersulfone (PES)	PES-0.45	0.45 μm	60%	130-150 μm
Polypropylene (PP)	PP-0.45	0.45 μm	40%	190 μm
	PP-Celgard	N/A	N/A	~250 μm
Zirfon UTP220 * Composite materials of zirconia and polysulfone	Zirfon-UTP220	0.45μm	50%	220 μm

Supplementary Table 4 | Comparisons of operation duration for CO_2R/COR to C_{2+} products. The system presented in this work is among one of the longest demonstrations in current literature operated at industrially relevant current densities. Rows in shades represent systems that suffer CO_2 loss.

Catalyst	Current Density	FE_{C2+}	Reactant	Stability	Reference
Graphite/C/Cu/PTFE	85 mA cm^{-2}	83%	CO_2-Alkaline flow cell	150 hrs	[12]
Molecule-tuned Cu	120 mA cm^{-2}	~80%	CO_2-Neutral MEA	180 hrs	[13]
Cu/PTFE	172 mA cm^{-2}	80%	CO_2-Netural MEA	157 hrs	[14]
Cu/PTFE	100 mA cm^{-2}	~70%	CO_2-Acidic MEA	200 hrs	[15]
Graphite/C/$Cu_{0.9}Zn_{0.1}$	150 mA cm^{-2}	82%	CO_2-Acidic flow cell	150 hrs	[16]
CG-Cu	100 mA cm^{-2}	~80%	CO_2-Acidic flow cell	155 hrs	[17]
Ag-Ru-Cu	300 mA cm^{-2}	93%	CO	100 hrs	[18]
CuO	200 mA cm^{-2}	90%	CO	110 hrs	[19]
Cu-Ag/DA	100 mA cm^{-2}	90%	CO	800 hrs	[20]
Cu-CCBH	100 mA cm^{-2}	83%	CO	200 hrs	[9]
Cu-NN	200 mA cm^{-2}	86%	CO	100 hrs	[21]
Cu	**200 mA cm^{-2}**	**96%**	**CO**	**250 hrs**	**This work**

Supplementary Table 5 | Comparisons of COR performance.

Catalyst	Cell Voltage	FE_{C2+}	EE_{C2+}	SPCE	Stability	Reference
Ag-Ru-Cu	-2.65 V @ 300 mA cm^{-2}	93%	37%	85%	100 hrs	[18]
CuO	-2.18 V @ 200 mA cm^{-2}	90%	43%	N/A	110 hrs	[19]
Cu-Ag/DA	-2.60 V @ 100 mA cm^{-2}	90%	~34%	71%	800 hrs	[20]
Ag-Cu_2O	-2.50 V @ 100 mA cm^{-2}	90%	~35%	N/A	18 hrs	[22]
Cu	-2.32 V @ 140 mA cm^{-2}	71%	29%	43%	24 hrs	[23]
Cu:Py:SSC	-2.41 V @ 120 mA cm^{-2}	76%	32%	27%	110 hrs	[24]
Cu-CCBH	-2.23 V @ 240 mA cm^{-2}	87%	40%	95%	200 hrs	[9]
OD-Cu	-3.25 V @ 500 mA cm^{-2}	90%	~27%	N/A	1 hr	[25]
Cu-NN	-2.30 V @ 180 mA cm^{-2}	86%	40%	~50%	100 hrs	[21]
Cu	**-1.95 V @ 200 mA cm^{-2}**	**96%**	**51%**	**53%**	**250 hours**	**This work**
Cu	**-1.95 V @ 200 mA cm^{-2}**	**91%**	**48%**	**97%**	**N/A**	**This work**

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